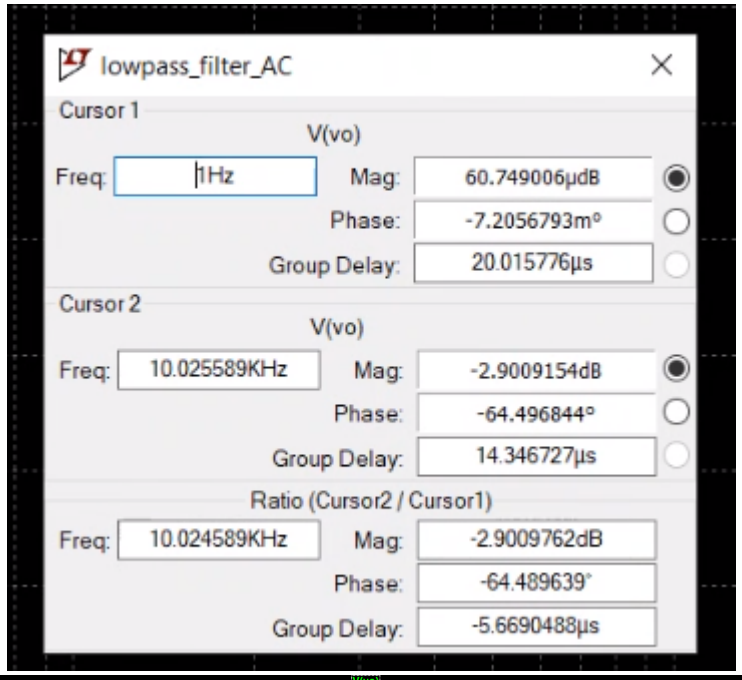
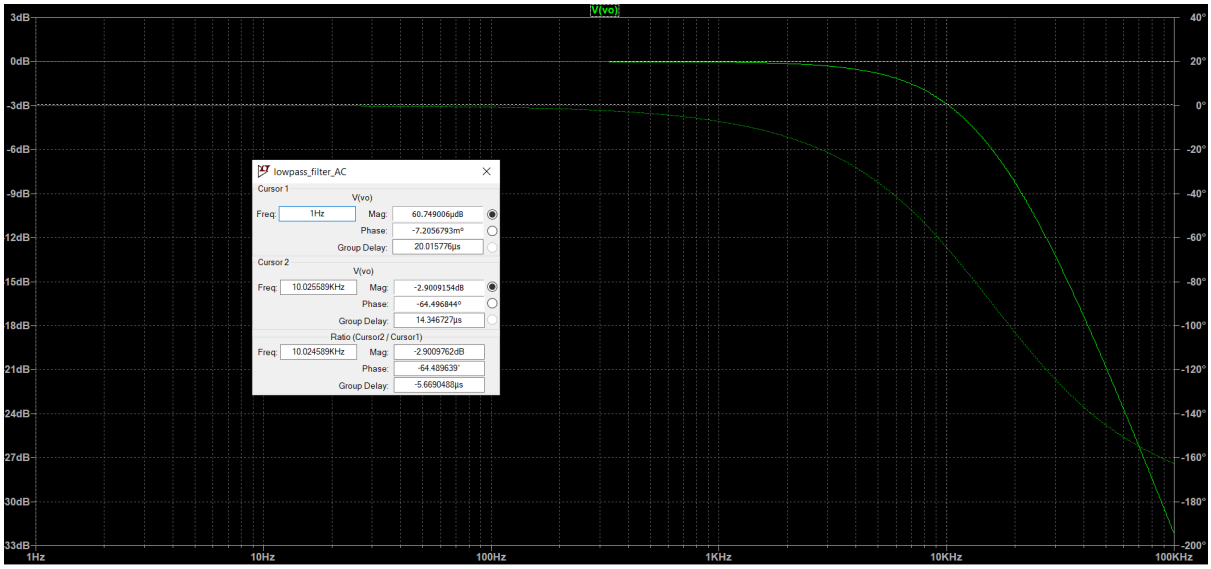
Laborator 6 EEA

3.1. Filtrul Trece-Jos realizat cu Amplificator operațional

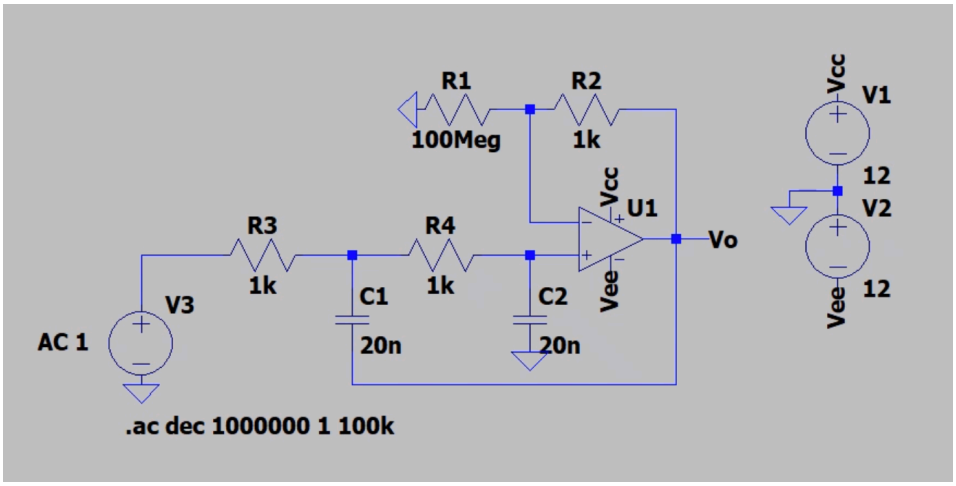
1. Frecvența caracteristică a filtrului este de 10 kHz.





2. Am ales să dublăm capacitatea condensatoarelor C1 și C2.

   Frecvența caracteristică a filtrului este de 5.1 kHz (se înjumătățește).



lowpass_filter_AC

Cursor 1

V(vo)

Freq:	1Hz	Mag:	60.646192µdB
		Phase:	-14.405618m°
		Group Delay:	40.015605µs

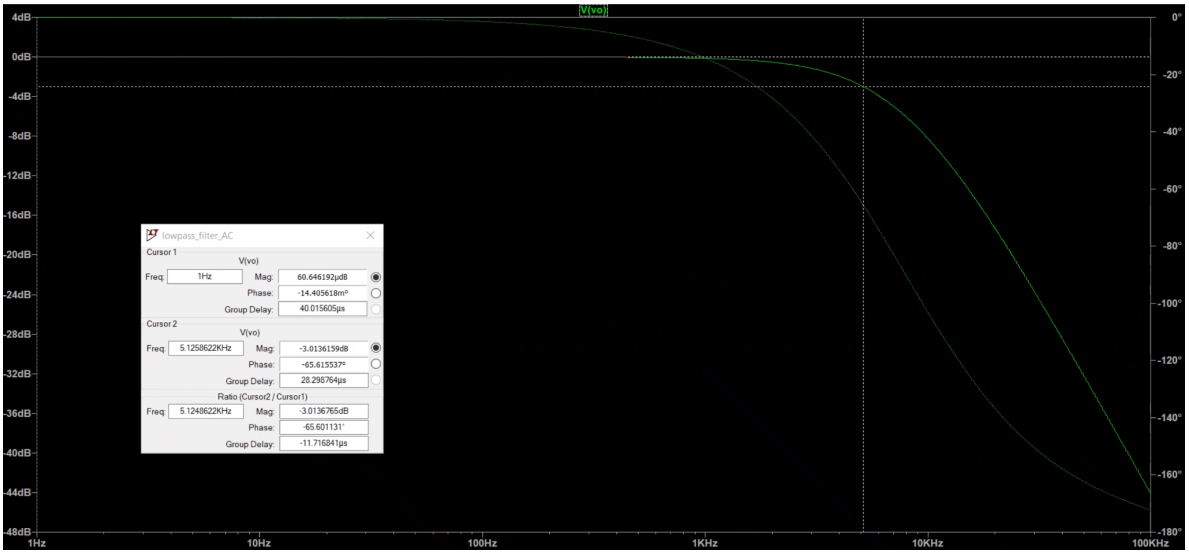
Cursor 2

V(vo)

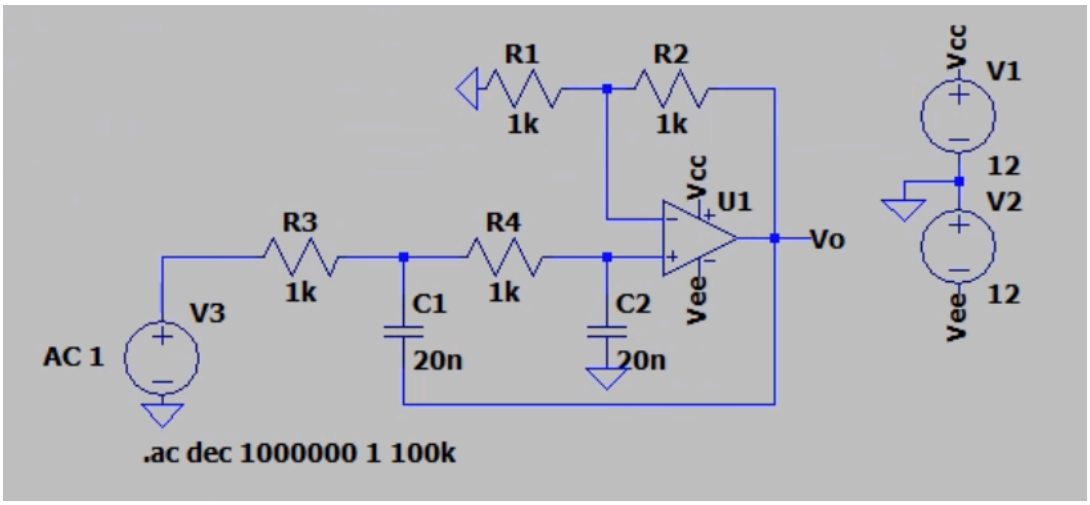
Freq:	5.1258622KHz	Mag:	-3.0136159dB
		Phase:	-65.615537°
		Group Delay:	28.298764µs

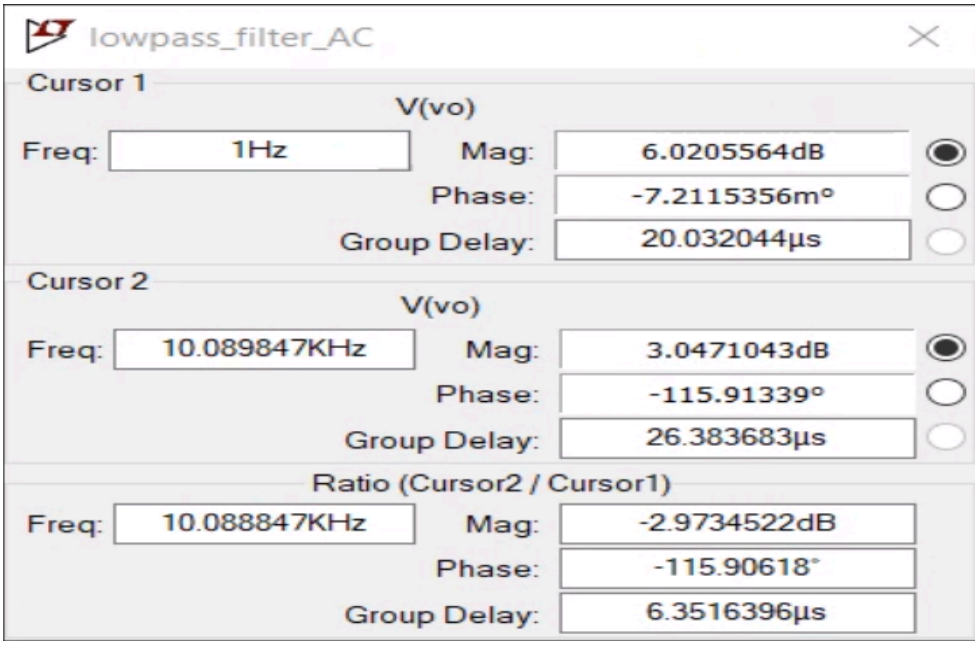
Ratio (Cursor2 / Cursor1)

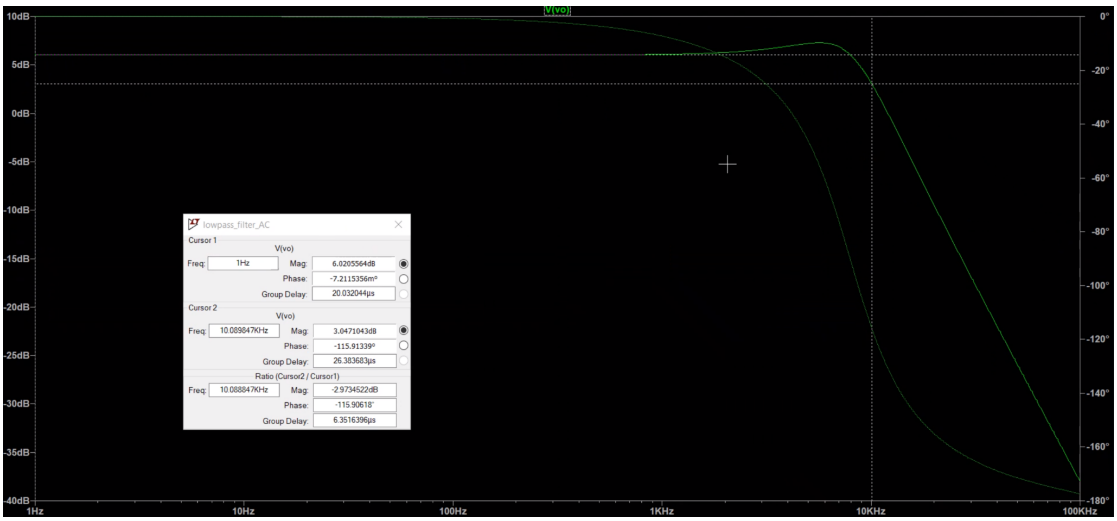
Freq:	5.1248622KHz	Mag:	-3.0136765dB
		Phase:	-65.601131°
		Group Delay:	-11.716841µs



3. Semnalul filtrului are o ușoară creștere (un spike) înainte de frecvența caracteristică, după care semnalul se st***abilizează.

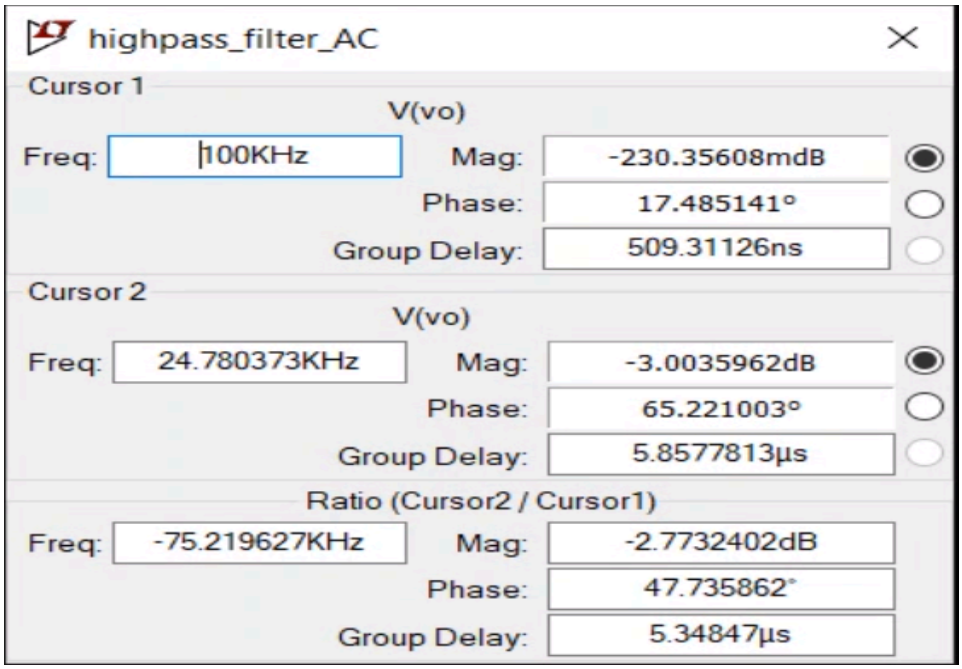


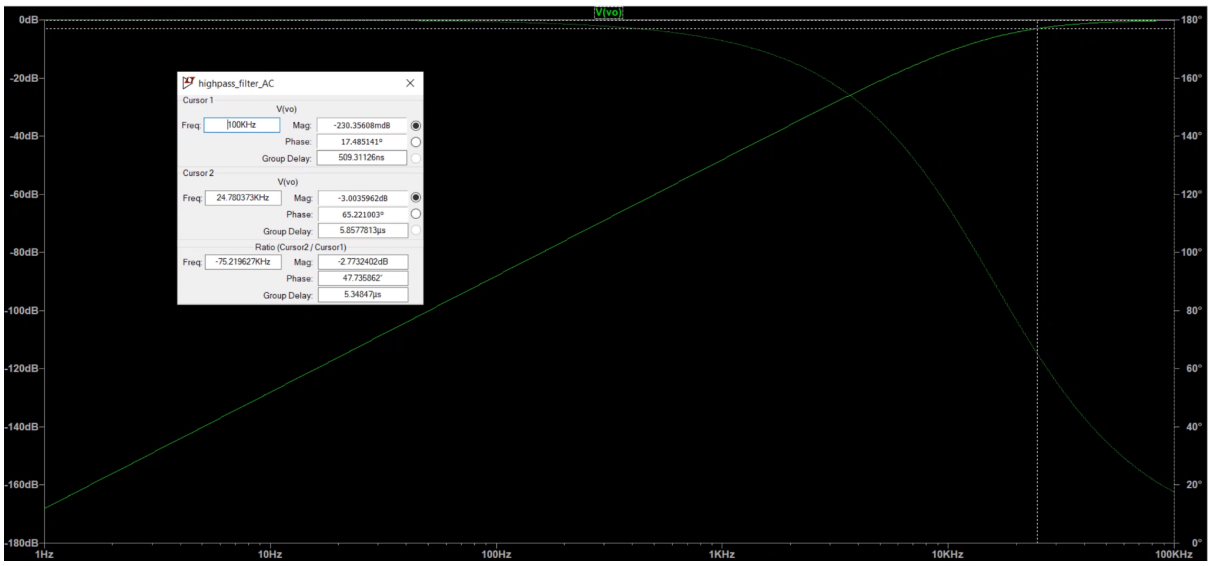




3.2. Filtrul Trece-Sus realizat cu Amplificator operațional

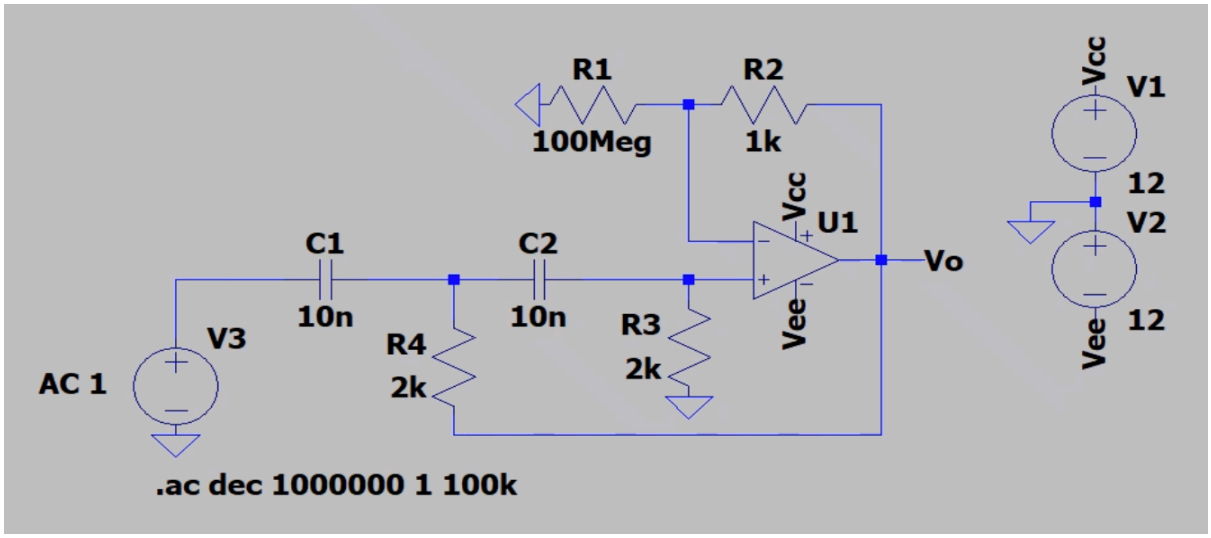
1. Frecvența caracteristică a filtrului este de 24.7 kHz.





2. Am dublat rezistențele filtrului (R3 și R4).

   Frecvența este 12.3 kHz (s-a înjumătățit).



highpass_filter_AC

Cursor 1

V(vo)

Freq:	100KHz	Mag:	-61.94309mdB
		Phase:	8.5189473°
		Group Delay:	267.44514ns

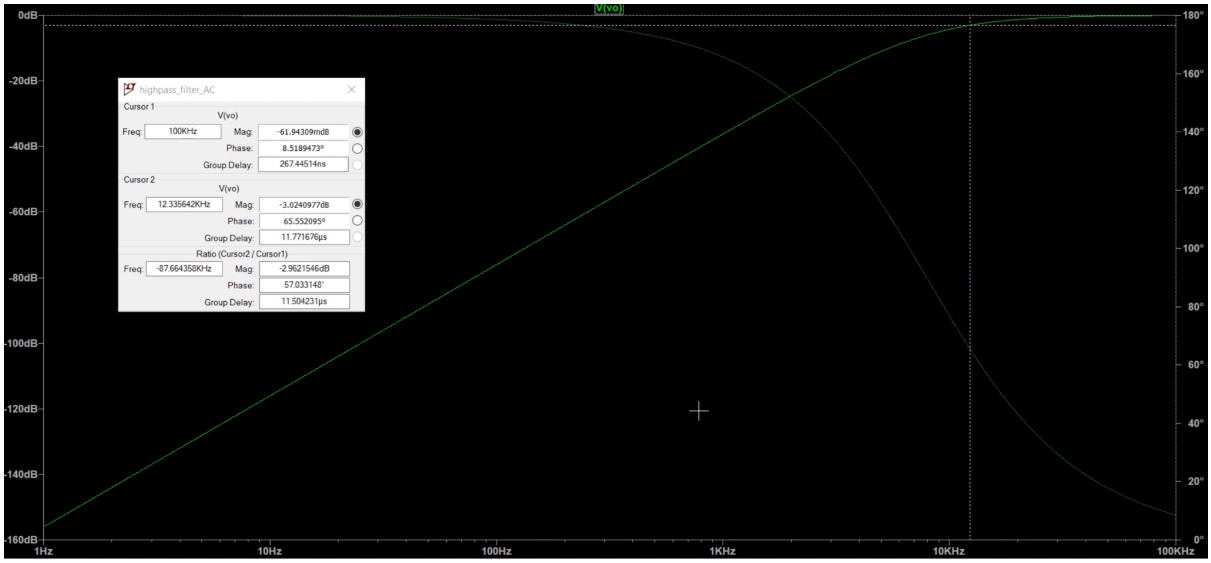
Cursor 2

V(vo)

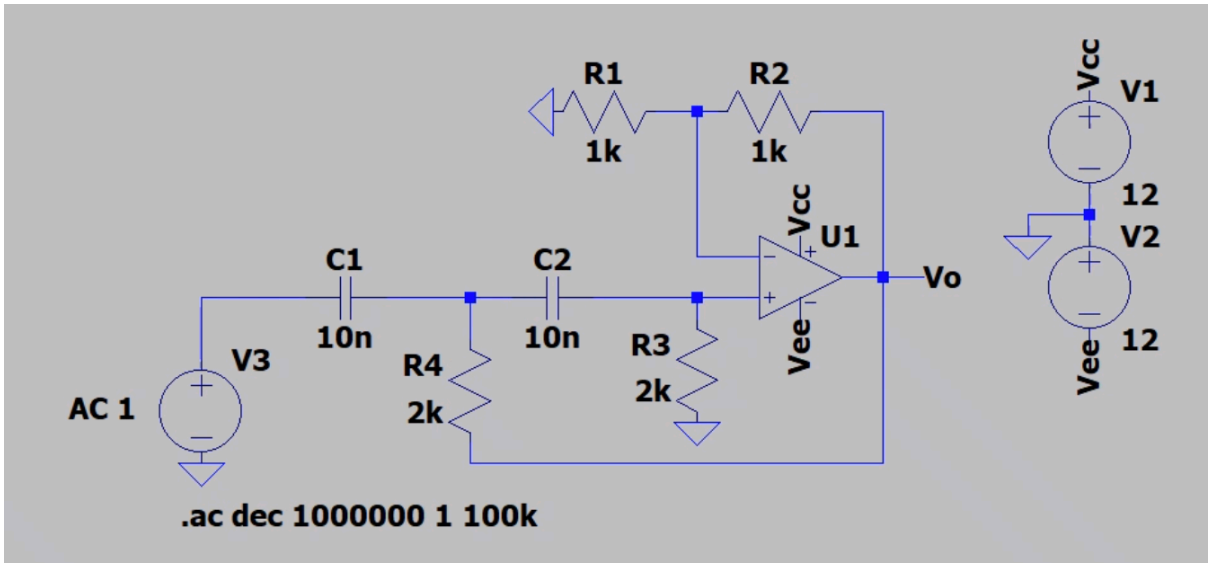
Freq:	12.335642KHz	Mag:	-3.0240977dB
		Phase:	65.552095°
		Group Delay:	11.771676µs

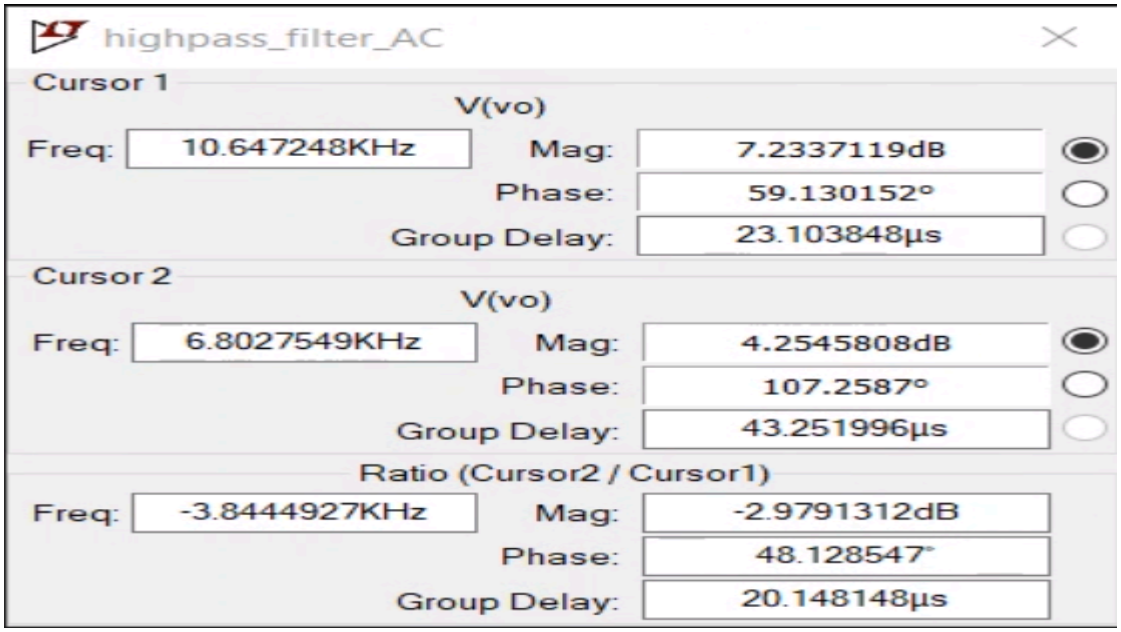
Ratio (Cursor2 / Cursor1)

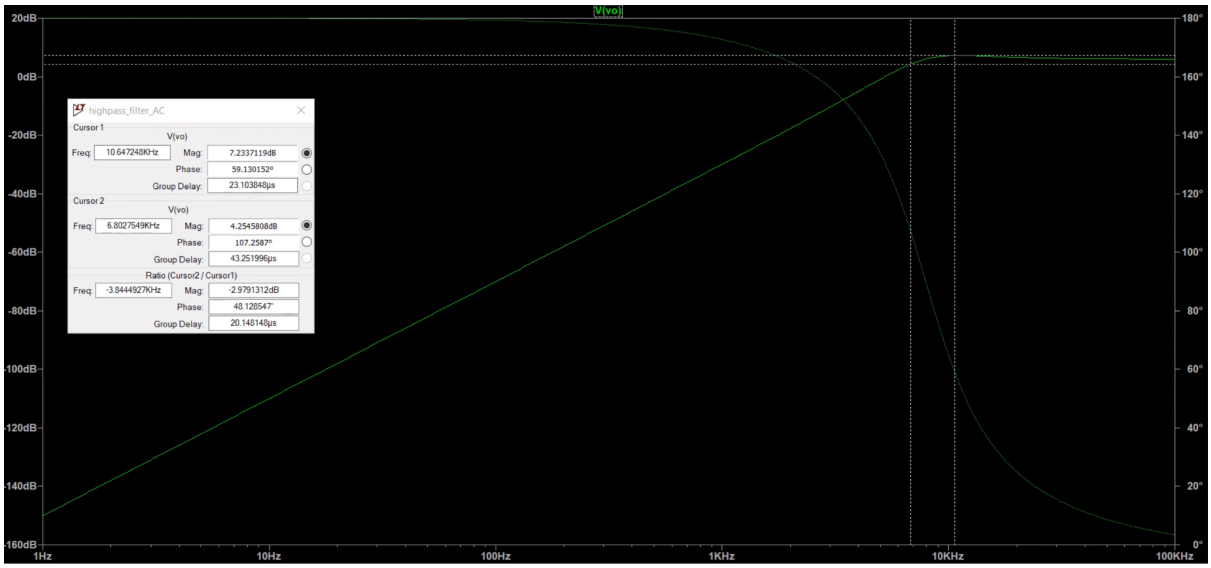
Freq:	-87.664358KHz	Mag:	-2.9621546dB
		Phase:	57.033148°
		Group Delay:	11.504231µs



3. Frecvența caracteristică este de 6.8 kHz. De la această frecvență, semnalul crește până la valoarea maximă, după care este într-o ușoară scădere.

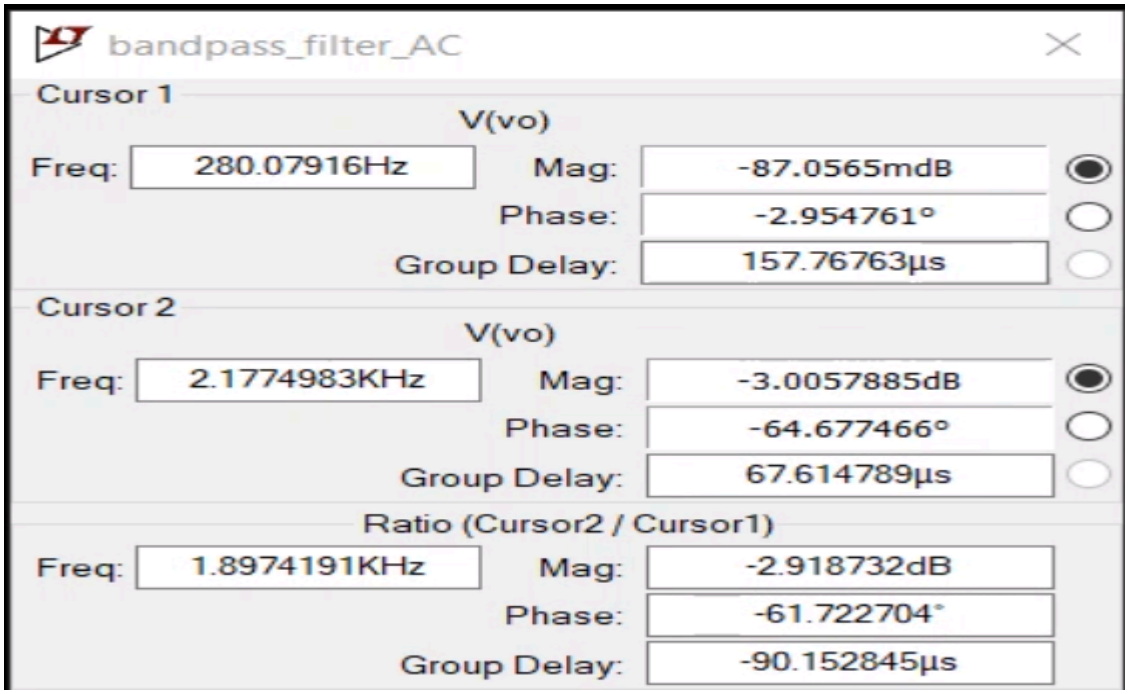


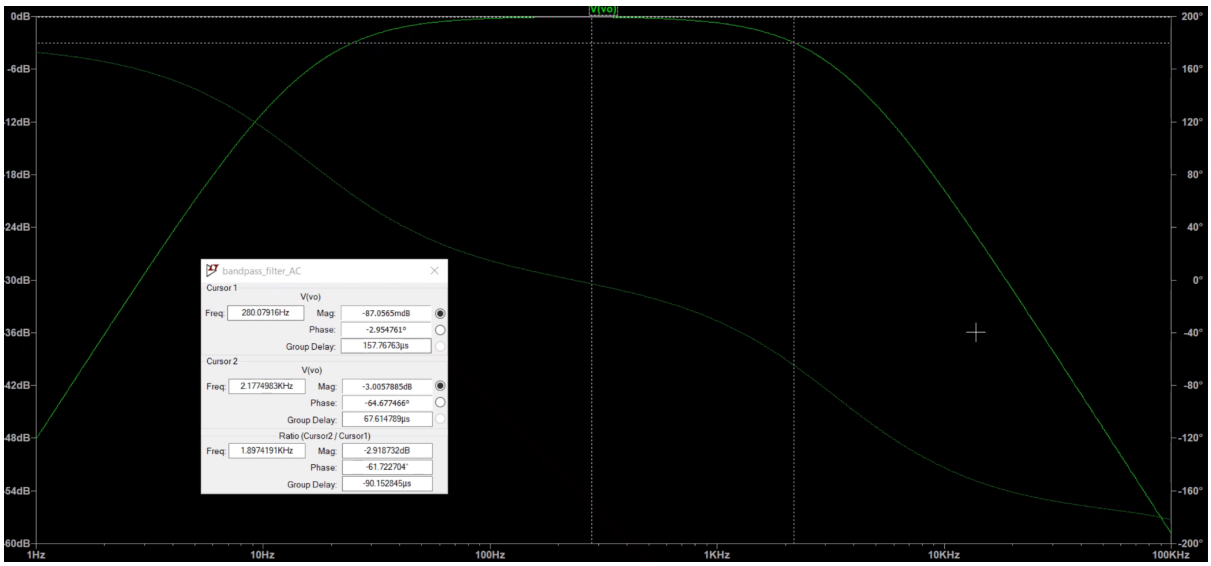




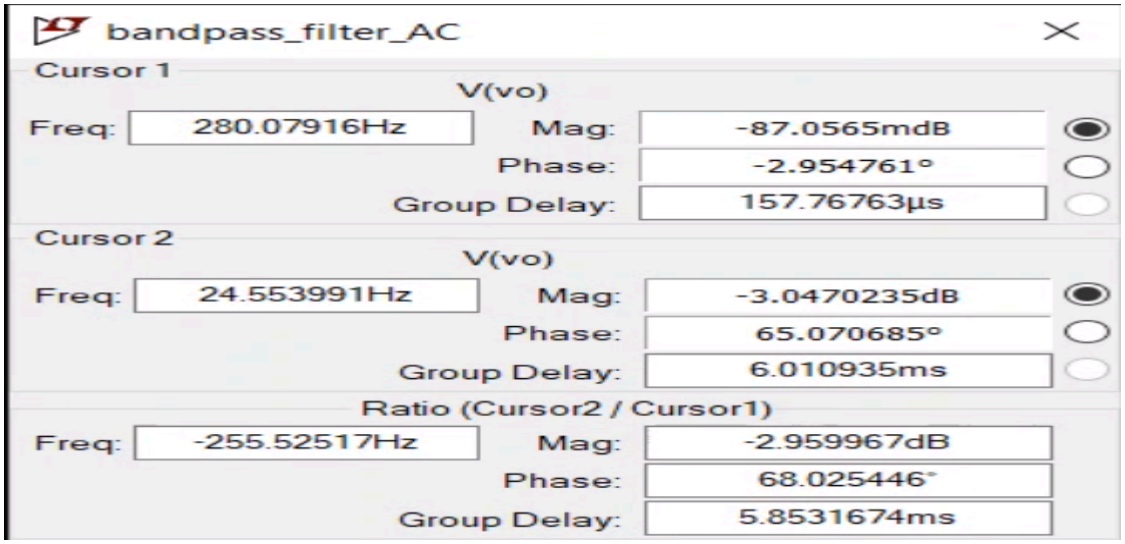
3.3. Filtrul Trece-Bandă realizat cu Amplificator operațional

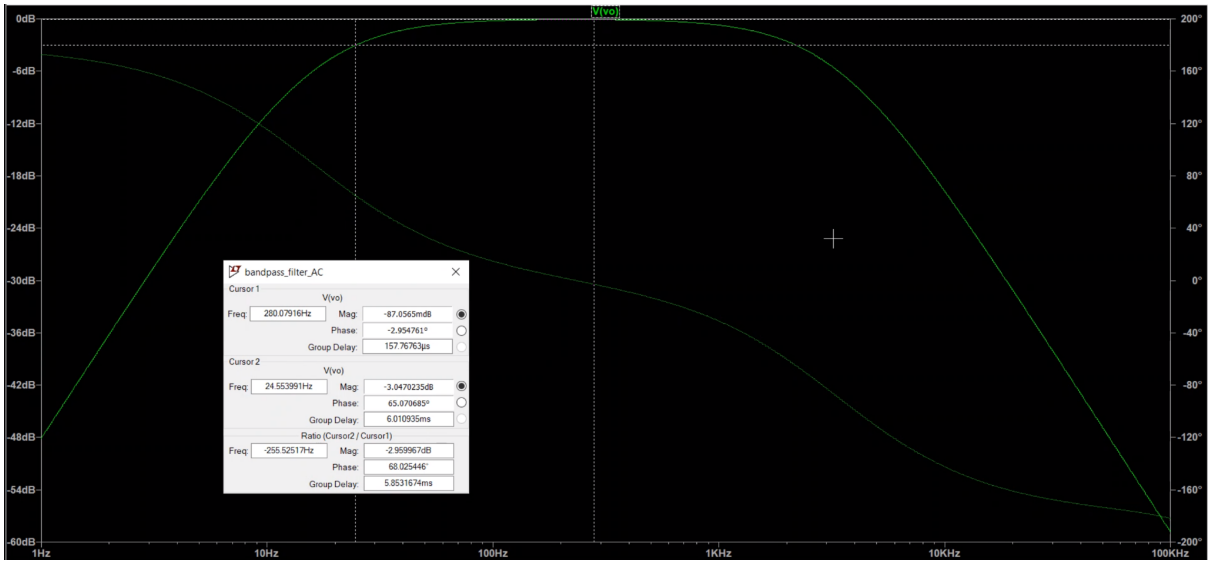
1. Frecvența maximă a benzii este de 2.2 kHz.



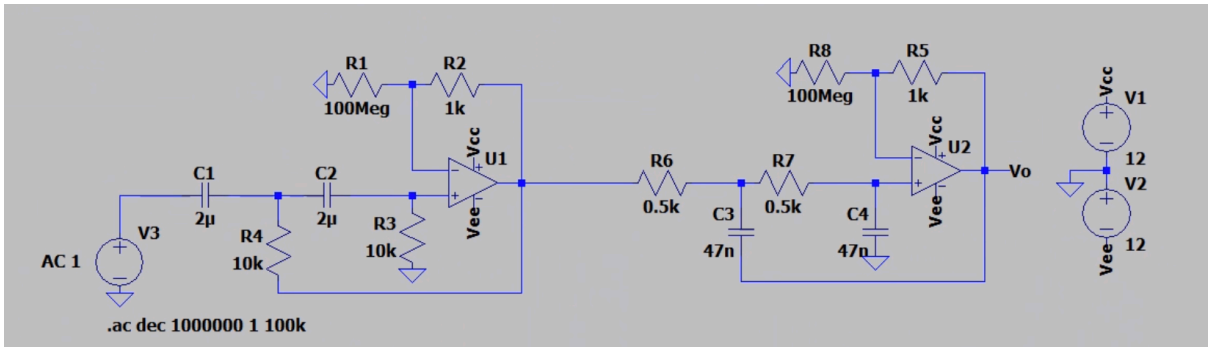


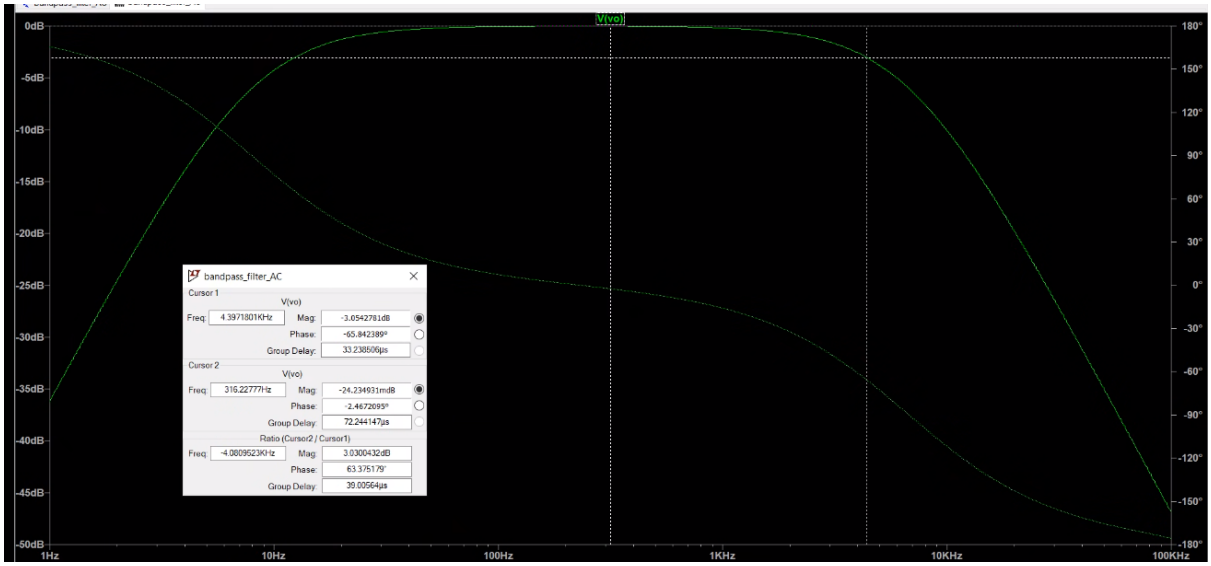
2. Frecvența minimă a benzii de trecere este 24 Hz.

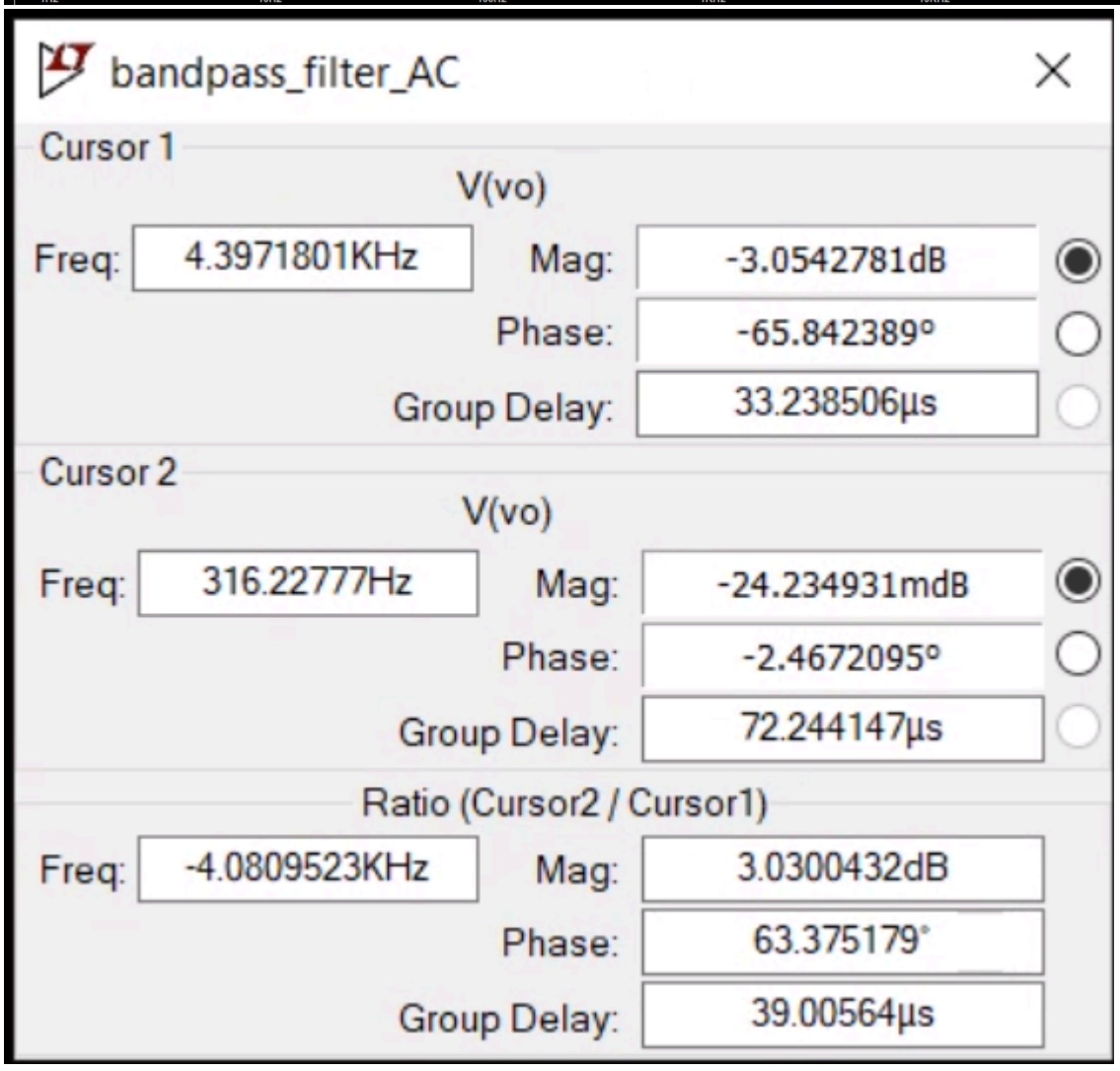


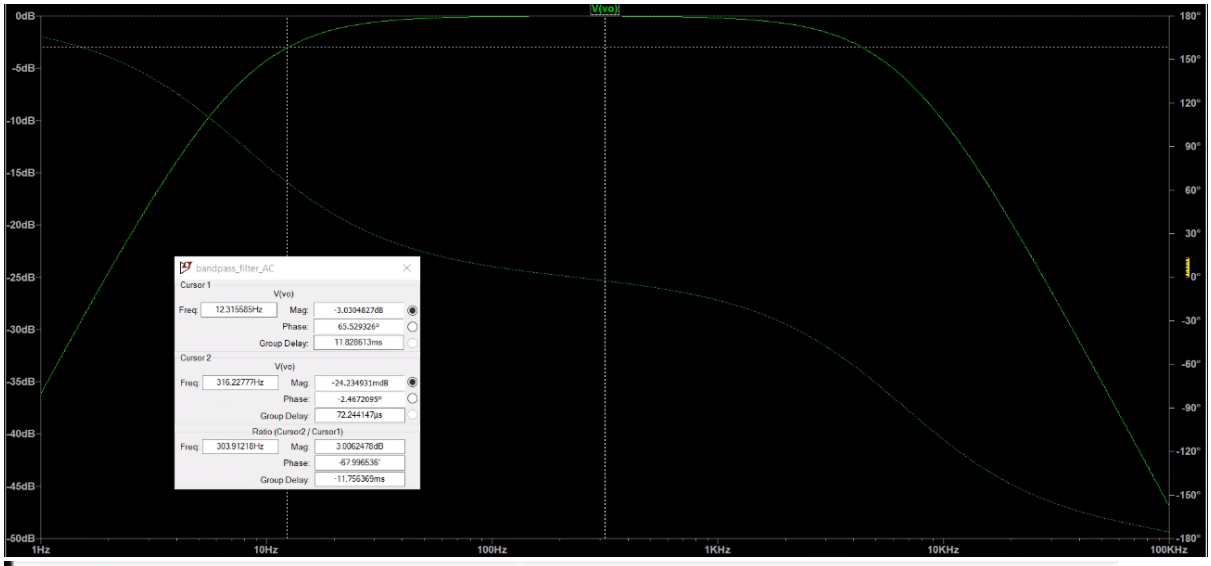


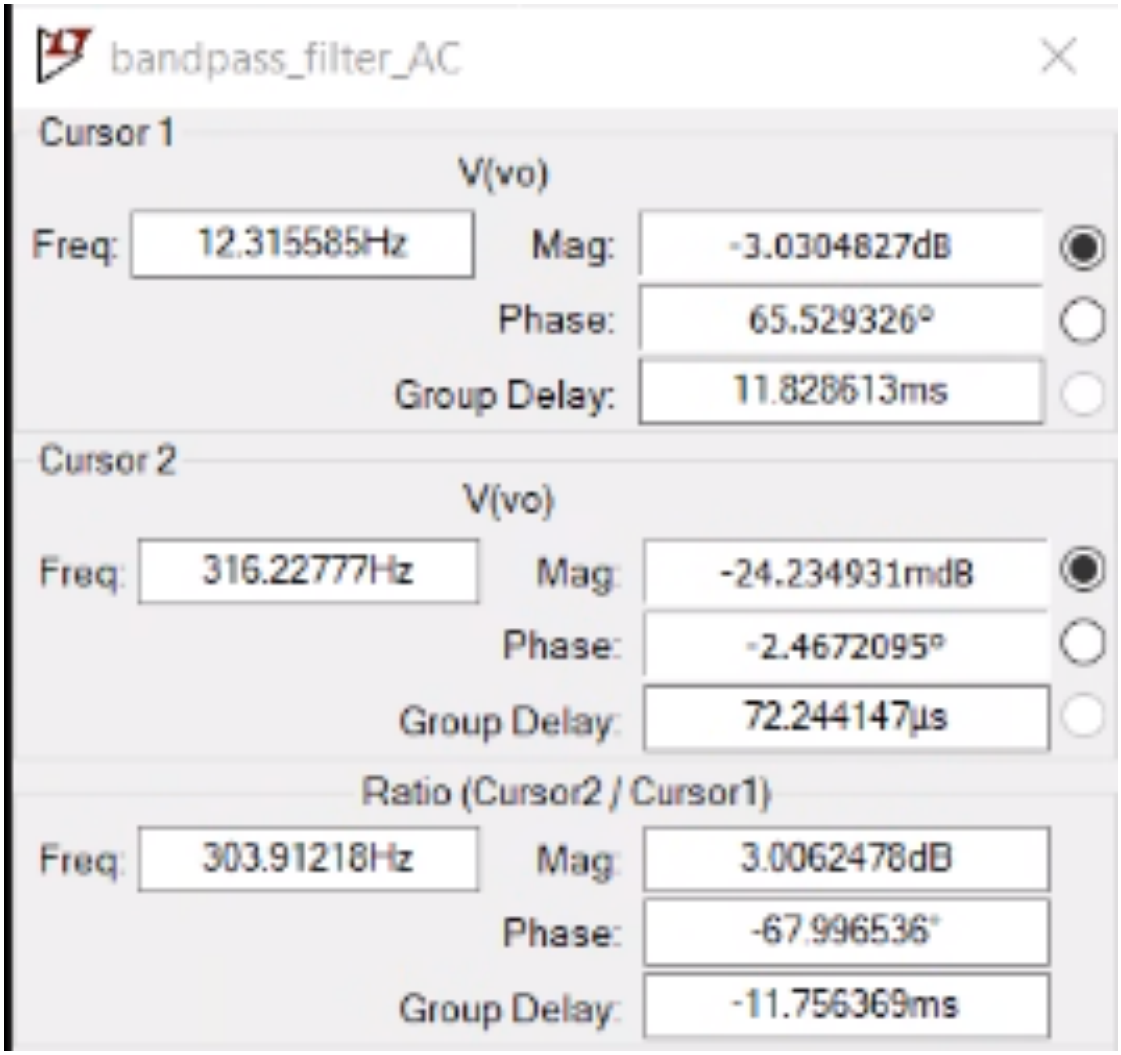
3. Filtrul Trece-Bandă este aproximativ format dintr-un filtru trece-sus și un filtru trece-jos. Astfel, am înjumătățit valoarea rezistențelor la trece-jos și la trece-sus am dublat valoarea condensatoarelor.



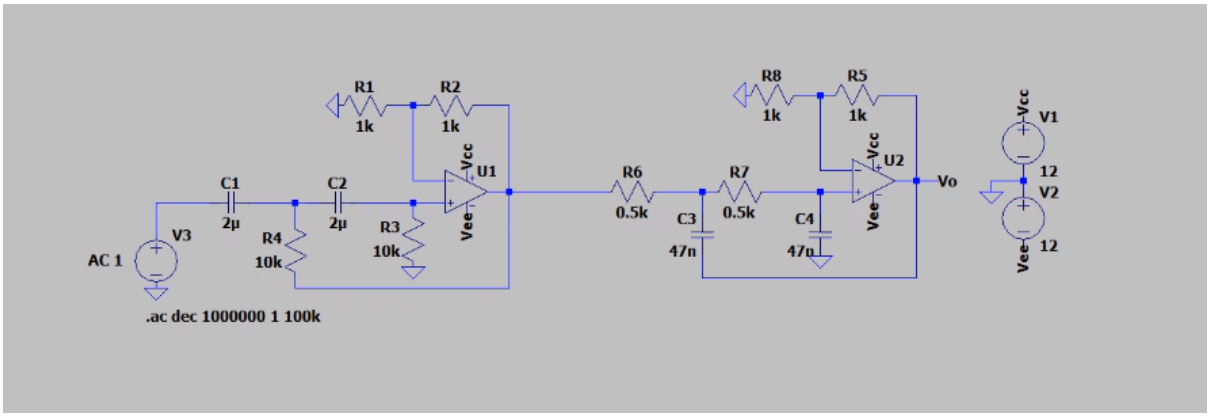

V(vo)
0dB
-5dB
-10dB
-15dB
-20dB
-25dB
-30dB
-35dB
-40dB
-45dB
-50dB
180°
150°
120°
90°
60°
30°
0°
-30°
-60°
-90°
-120°
-150°
-180°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
bandpass_filter_AC
Cursor 1
V(vo)
Freq: 4.3971801KHz
Mag: -3.0542781dB
Phase: -65.842389°
Group Delay: 33.238506µs
Cursor 2
V(vo)
Freq: 316.22777Hz
Mag: -24.234931mdB
Phase: -2.4672095°
Group Delay: 72.244147µs
Ratio (Cursor2 / Cursor1)
Freq: -4.0809523KHz
Mag: 3.0300432dB
Phase: 63.375179°
Group Delay: 39.00564µs

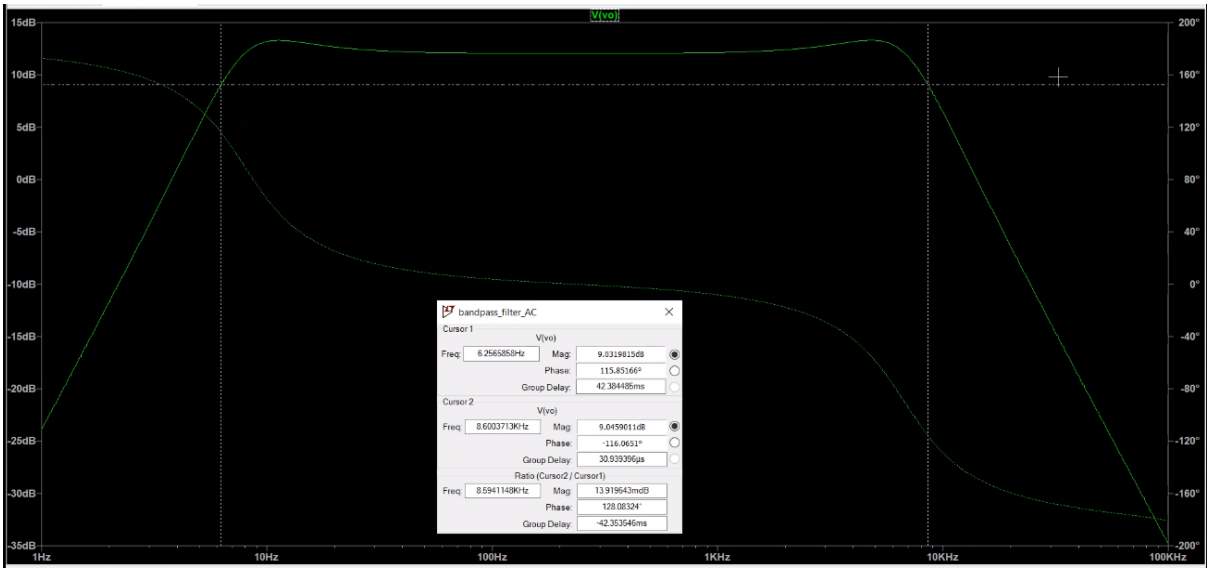

bandpass_filter_AC
Cursor 1
V(vo)
Freq: 4.3971801KHz
Mag: -3.0542781dB
Phase: -65.842389°
Group Delay: 33.238506µs
Cursor 2
V(vo)
Freq: 316.22777Hz
Mag: -24.234931mdB
Phase: -2.4672095°
Group Delay: 72.244147µs
Ratio (Cursor2 / Cursor1)
Freq: -4.0809523KHz
Mag: 3.0300432dB
Phase: 63.375179°
Group Delay: 39.00564µs

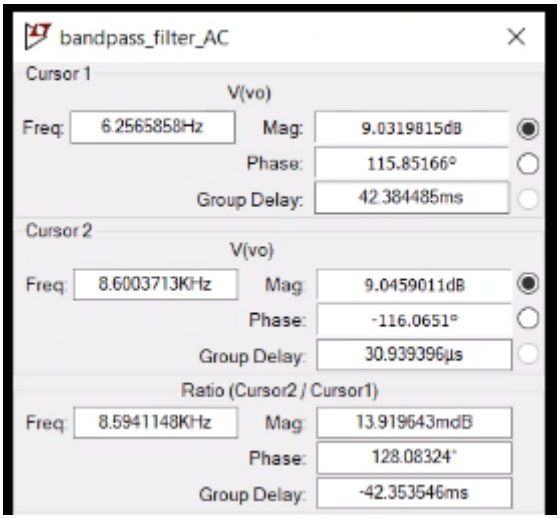




4. Am echivalat toate rezistențele neinversoare cu 1 koHm pentru a obține o amplificare 2.

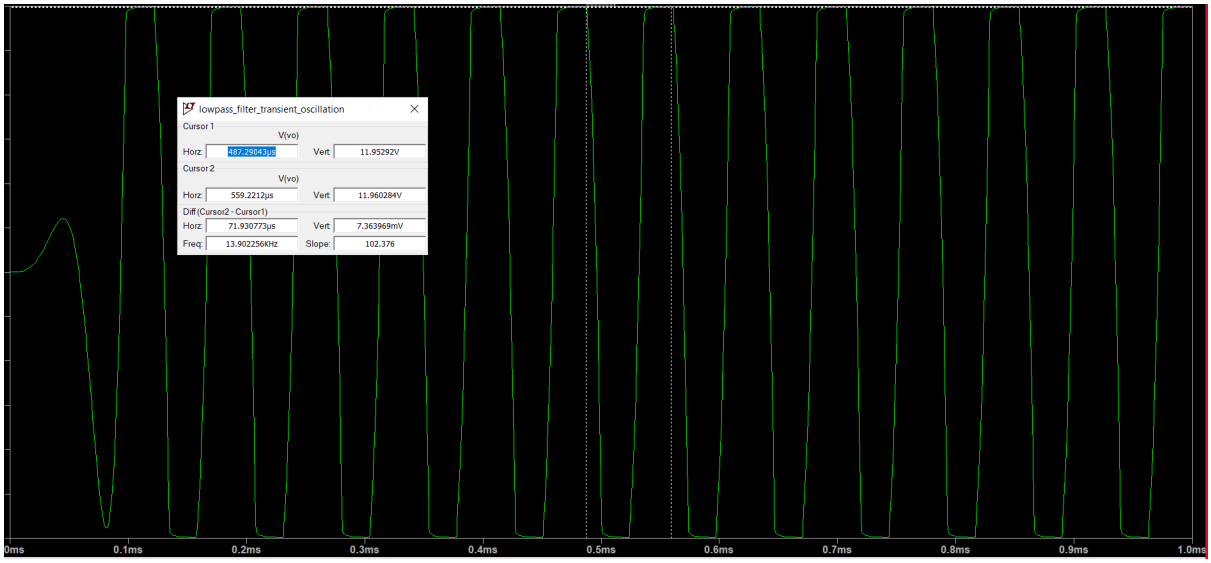






În apropierea frecvenței caracteristice, avem atât pentru frecvența maximă și minimă câte un spike, după care se stabilizează.

3.4 Rezistentele care realizeaza amplificarea in circuit sunt R1 si R2, amplificarea fiind A = 1 + R2/R1 = 4


lowpass_filter_transient_oscillation
Cursor 1
V(vo)
Horz: 467.2904µs
Vert: 11.95292V
Cursor 2
V(vo)
Horz: 559.2212µs
Vert: 11.960284V
Diff (Cursor2 - Cursor1)
Horz: 71.930773µs
Vert: 7.363969mV
Freq: 13.902256KHz
Slope: 102.376
0ms
0.1ms
0.2ms
0.3ms
0.4ms
0.5ms
0.6ms
0.7ms
0.8ms
0.9ms
1.0ms